

CHI WEIQI

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EDUCATION

THE UNIVERSITY OF TOKYO	TOKYO, JAPAN
Ph.D. in Information Engineering	Apr. 2024 – Present
TOKYO INSTITUTE OF TECHNOLOGY	TOKYO, JAPAN
Master of Engineering, Wireless Communication Engineering, Electromagnetics	Sep. 2020 – Sep. 2022
TIANJIN UNIVERSITY OF FINANCE AND ECONOMICS	TIANJIN, CHINA
Bachelor of Engineering, Information and Engineering	Sep. 2016 – Jun. 2020

RESEARCH EXPERIENCE

Dynamic mmWave Blockage Prediction in RSU assisted ITS Apr. 2024 – current

Doctoral Research

- **Abstract:** This study focuses on the millimeter-wave (mmWave) enabled vehicular network while addressing the significant challenge of link quality degradation due to blockages. A novel blockage prediction utilizing the future path of vehicles is proposed, a blockage-aware ITSs supported by the Road-Side Units (RSU) is validated through recent advanced Vehicular ad-hoc network (VANET) simulation platforms.
- **Key Contributions:**
 - a) Proposed system structure for future path sharing based mmWave blockage prediction;
 - b) First Fresnel zone-based blockage determination algorithm validation;
 - c) Integrated SUMO and Wireless Insite (WI) simulation for real-world 3D map (Unity & Blender) based advanced V2X communication environment modeling;

A Blockage-Aware mmWave V2X Connectivity Analysis based on MCM Nov. 2024 – Jan. 2025

Visiting Researcher in Technische Hochschule Ingolstadt, Germany

- **Abstract:** This research proposes a blockage-aware communication scheme for mmWave V2X networks using Maneuver Coordination Messages (MCM) and Future Dynamic Map (FDM) to enhance RSU-assisted ITSs for predictive blockage mitigation.
- **Key Contributions:**
 - a) Proposed a FDM-based data structure for advanced V2X service development;
 - b) Proposed an MCM-based blockage prediction scheme for V2X performance enhancement;
 - c) VANET simulation (Artery, OMNeT++, SUMO) with Artery core development team;

Measurement and Prediction of Attenuation caused by vegetation on mmWave Sep. 2020 – Jul. 2022

Master Research & Joint Research with Rakuten Mobile, Inc.

- **Abstract:** Measure and predict the static shadowing effect caused by a single tree at 25GHz/28GHz by designing outdoor experiment and simulation of different models.
- **Key Contributions:**
 - a) Designed the outdoor measurement for both 5G base station and portable mmWave base stations (Area tester, Spectrum analyzer);
 - b) **Indoor pre-experiments** under Line-of-Sight (LOS) scenario;
 - c) Double knife edge diffraction modeling (DKED model) under the measurement scenario;
 - d) ITU-R tree fading model simulation by exploiting the tree dimension data captured by the laser scanner;
 - e) WI simulation by building 3D measurement environment and tree model;
 - f) Laser scanning, point cloud fusion, ranging, antenna pattern analyzing;
 - g) Research discussions with Rakuten Mobile group and oral presentations on Super Smart Society (SSS) student conferences;

PUBLICATION

Internation conference

[1] **Weiqi Chi**, Jin Nakazato, Tomoki Murakami, Manabu Tsukada, "V2I Blockage Modeling and Performance Evaluation for Connected Autonomous Vehicle", The IEEE 99th Vehicular Technology Conference (VTC2024-Spring), Singapore, Jun. 2024. (Peer Reviewed & Oral Presentation)

[2] **Weiqi Chi**, N. Keerativoranan, J. Takada, "Outdoor mmWave Band Tx Localization Using Scattering Path under Spectrum Sharing Scenario," 13th AOTULE (Asia-Oceania Top University League on Engineering) Student Conference, Daejeon, Korea, Nov. 2021 (Oral Presentation)

Domestic conference

[1] **Weiqi Chi**, N. Keerativoranan, J. Takada, "Outdoor Measurement of Millimeter-Wave Propagating Through Vegetation at 28GHz Band," MISW Conference, Tokyo, Japan, Aug. 2022 (Oral Presentation & Chairperson).

[2] **Weiqi Chi**, Jin Nakazato, Tomoki Murakami, Manabu Tsukada, "ミリ波帯における遮蔽対応型 V2I 通信と解決策について", in IEICE 無線通信システム研究会, Kyoto Institute of Technology, Mar. 2025. (Oral Presentation)

Poster

[1] **Weiqi Chi** and M. Tsukada, "Blockage-aware Multi-armed Bandits for Proactive Handover in mmWave Vehicular Networks," in Cooperative Interactive Vehicles (CIV) Summer School 2025, Geneva, Switzerland, 2025.

EMPLOYMENT HISTORY

General Test Systems Inc., Shenzhen, China

Feb. 2023 – Dec. 2023

Application engineer

- **Introduction:** General test focus on providing end-to-end wireless communication Over-the-air (OTA) testing solutions with various production lines.
- **Work specification:**
 - a) OTA testing for Internet Connected Vehicles (ICVs), Anechoic chamber experience;
 - b) Hardware knowledge of OTA testing system
 - c) Operational experience of testing instrumentation (VNA, Radio Communication Analyzer, Area Tester);

Rakuten Mobile Inc. R&D group, drone application development, Japan

Sep. 2022 – Sep. 2022

Internship

- **Introduction:** This is a 2-week preliminary research investigation for the R&D team in Rakuten Mobile for their upcoming drone development project.
- **Work specification:** Drone application development based on DJI Mobile SDK on Android platform

SKILL

- **Programming skills:** C, C++, Python, MATLAB, OMNeT++ (Ned language)
- **Development Tools:** Git, Docker
- **Visualization:** Unity, Blender
- **Simulation & Modeling:** WI, OMNeT++, SUMO, Veins & Artery (V2X simulation platform)
- **Language:** English (Native) , Japanese (JLPT N1) , Chinese (Native), German (Beginner)
- **Interests:** Piano, Music, Swimming, Boxing, Bouldering

SCHOLARSHIP

- JST Support for Pioneering Research Initiated by the Next Generation (SPRING) Program, The University of Tokyo "Advanced Human Resource Development Leading Green Transformation (GX)" (**SPRING GX**) (Apr.2024-Apr.2027)
- Project student member of Tokyo Institute of Technology **Supper Smart Society (SSS)** (Sep.2021-Sep.2022)
- **First-class scholarship & Merit student** of Tianjin University of Finance and Economy (Jun. 2018)

AWARD

- **Young Researcher's Encouragement Award**, IEEE VTS Tokyo/Japan Chapter 2024 (Jun. 2024)
- **The Best Oral presenter**, MISW (Multidisciplinary International Student Workshop) Conference (Aug. 2021)
- **First Award**, CUMCM (Contemporary Undergraduate Mathematical Contest in Modeling), Tianjin, (2018)
- **Third Award**, **Japanese speech contest** at the 11th Western Culture Festival. (2018)

- ***Excellent project award***, Tianjin University of Finance and Economics, 2017 SRT project, "OFDM peak-to-average ratio suppression and power amplifier improvements", (2017)
- *Second Award*, the 22th “China Daily 21st Century · Coca-Cola Cup” College **English Speaking Competition** (Tianjin Region), (2016)